

CHAPTER 1

INTRODUCTION

A maze-solving robot is an intelligent autonomous mobile system designed to analyze and navigate through a given maze while finding the most efficient path from a defined starting point to the final destination without any human intervention. These robots operate using various onboard sensors such as ultrasonic sensors, infrared proximity sensors, line sensors and occasionally encoders or gyroscopes to detect walls, measure distances, identify pathways and avoid obstacles. The collected sensor data is processed by a microcontroller, typically Arduino Nano or similar embedded platforms which interprets real-time input and makes navigation decisions. To accomplish this, the robot uses logical decision-making and algorithms such as the Right-Hand Rule, Left-Hand Rule, Flood-Fill or Wall-Follower techniques to evaluate possible routes and determine the shortest or most optimal path.

The development and implementation of maze-solving robots serve as a practical foundation for understanding key concepts in robotics, embedded systems, automation, artificial intelligence and control systems. Such projects play a vital role in bridging theoretical knowledge with real-world application by integrating hardware components like motors, wheels, microcontrollers and sensors with software elements including programming logic, path-planning algorithms and control strategies. Additionally the process encourages systematic problem-solving, critical thinking, error debugging and iterative improvement skills essential in engineering and research.

Maze-solving robotics also have significant real-world relevance in various industries. Their core principles are directly applied in robotic path-planning, autonomous vehicle navigation, warehouse automation, military scouting, search and rescue operations and intelligent transportation systems.

AIM

The aim of this project is to design and develop an autonomous maze-solving robot using the Arduino Uno microcontroller, capable of detecting walls and navigating through a maze using appropriate sensors and logic-based pathfinding algorithms. The robot will analyze its environment, make real-time decisions, and find an efficient path from the start point to the goal without human intervention, thereby demonstrating fundamental concepts of robotics, embedded systems, and algorithmic problem-solving.

PROBLEM STATEMENT

In many real-world applications such as disaster response, warehouse management and autonomous transportation robots must be able to explore and navigate unknown environments without human control. Developing such autonomous systems requires reliable sensing, intelligent navigation algorithms and real-time decision-making capabilities which are essential for operating safely and efficiently in unpredictable conditions.

However, building low-cost and accessible autonomous robots remains a challenge especially for students and beginners. Limited experience with sensor interfacing, algorithm implementation and hardware–software integration creates difficulties in learning and prototyping. Existing advanced robotics platforms are often expensive and complex making them unsuitable for foundational learning. Therefore, there is a need for a simple and affordable educational platform that helps learners understand and implement autonomous navigation concepts in a practical way.

OBJECTIVE

The objective of this project is to design and develop an autonomous maze-solving robot using an Arduino Nano microcontroller. The robot will navigate a maze without human assistance by detecting walls and obstacles using ultrasonic or infrared sensors. It will use maze-solving algorithms such as the left-hand rule, right-hand rule, depth-first search, or flood-fill to determine the correct path. The system will process sensor data in real time and control the motors for accurate movement. The final goal is to build a low-cost, efficient, and educational robotics platform that helps students learn embedded systems, navigation algorithms, and intelligent control.

CHAPTER 2

LITERATURE SURVEY

AUTHOR	PAPER	SOURCE	FINDING
1.Ramesh R 2.Selvalakshmi 3.Naven Karthick M 4.Nikileshkumar S L 5.Cibivarshan S 6. Prasanth K	Design and Implementation of Autonomous Maze-Solving Robot based on an Enhanced Maze solving technique.	International Conference on Automation, Computing and Renewable Systems (ICACRS-2023)IEEE Xplore Part Number: CFP23CB5-ART; ISBN:979-8-3503-4023-5	A self-governing robot was designed to solve mazes using distance sensors and wheel encoders.The robot uses the flood-fill algorithm. That can work in places difficult or unsafe for humans.This approach is useful for robotic pathfinding and tracking applications.
1.Bindhu S 2.Tharun Gowda 3.Mr.Balaram M	Maze Solving Robot	International Research Journal of Modernization in Engineering Technology and Science e-ISSN: 2582-5208 Volume:06,Issue:06,June-2024	Maze-solving robots use sensors and algorithms for navigation. Proper calibration prevents damage and ensures accuracy.Algorithms manage movement when motors lack precision.
1.Rahul Kumar 2.Peni Jitoko 3.Sumeet Kumar 4.Krishneel Pillay	Maze solving robot with automated obstacle avoidance	International Symposium on Robotics and Intelligent Sensors, IRIS 2016, 17-20 December 2016, Tokyo, Japan	Autonomous MazeBot maps environment, avoids collisions, and uses a robotic arm to pick and place objects.

1.Sowmya P 2.Srinivasa G S 3.Vunnam Naga 4.Sahithi 5.Shivaraj Prakash Naik 6.Ashwal H U Shastry	A Maze Breakthrough Robot	MargShodhak 2023 IJNRD Volume 8, Issue 6 June 2023 ISSN: 2456-4184 IJNRD.ORG	The project shows that maze-solving robots rely on decision-making algorithms and sensors to navigate unknown paths. An Arduino-based autonomous robot successfully demonstrated intelligent movement without human control.
1.Rares Covaci 2.Gabriel Harja 3.Ioan Nascu	Autonomous Maze Solving Robot	978-1-7281-7166-1/20/\$31.00 2020 IEEE	The robot successfully explored and mapped the maze, detected paths using sensors, and solved it optimally on the second run. Bluetooth data transmission supported monitoring, and PI control improved navigation accuracy.

CHAPTER 3

METHODOLOGY

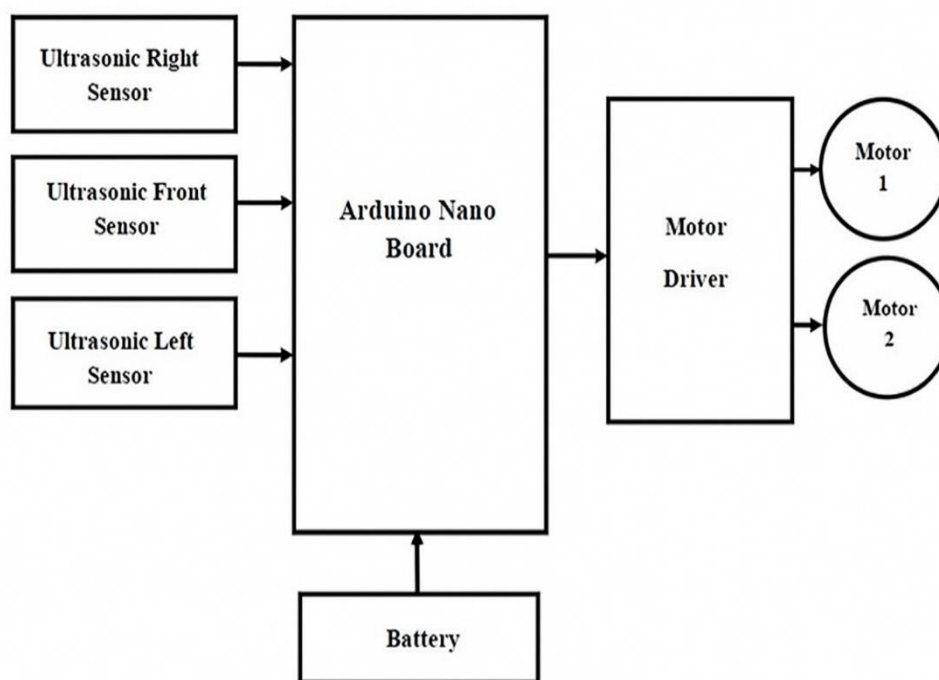


Fig 3.1 Basic Block Diagram

Block diagram description

The maze-solving robot operates using a set of ultrasonic sensors placed at the front, left, and right sides of the robot to continuously detect obstacles in the surrounding environment. These sensors send distance readings to the Arduino Nano microcontroller, which acts as the central processing unit. Based on the sensor inputs, the Arduino executes the programmed decision-making algorithm to determine the robot's movement direction. The processed control signals are then sent to the motor driver, which powers and regulates the two DC motors responsible for the robot's motion. The motor driver ensures safe current and voltage levels while allowing the robot to move forward, turn left, or turn right depending on the detected obstacles. The entire system is powered by a rechargeable battery that supplies energy to the Arduino board, sensors, and motors. This integrated setup enables the robot to autonomously navigate and solve the maze without human intervention.

Arduino Nano



Fig 3.2 Arduino Nano

The Arduino Nano is one of the most popular and compact development boards in the Arduino family, designed for ease of use in small-scale electronic and embedded projects. It is powered by the ATmega328P microcontroller, the same chip used in the Arduino Uno, but arranged on a much smaller board. It runs at a 16 MHz clock frequency, providing fast processing for various real-time applications. The Nano has 32 KB of flash memory for storing programs, 2 KB of SRAM for temporary data and 1 KB of EEPROM for permanent data storage.

The board provides 14 digital I/O pins, including 6 PWM (Pulse Width Modulation) pins that allow control of devices like motors and LEDs with varying intensity. It also includes 6 analog input pins that can read sensor values with 10-bit resolution. The Arduino Nano supports several communication interfaces such as UART (serial communication), SPI (for high-speed communication with modules like sensors or SD cards), and I2C (for connecting multiple devices using only two wires).

The Nano can be powered either through the Mini USB Type-B connector or via an external 7–12V power supply. It operates at a 5V logic level, though it can accept 3.3V inputs for certain components. One of its biggest advantages is its small form factor (45 x 18 mm), which makes it ideal for use on breadboards and compact circuit designs.

Additionally, the Arduino Nano is open-source, meaning its schematics and software are freely available for modification and learning. It is programmed using the Arduino IDE, which uses a simplified version of C/C++, making it beginner-friendly yet powerful enough for advanced users. Common applications of the Arduino Nano include robotics, automation systems, sensor-based projects, IoT (Internet of Things) devices and wearable electronics.

Features of Arduino NANO

- Microcontroller: ATmega328P
- Operating Voltage: 5V
- Input Voltage (recommended): 7–12V
- Clock Speed: 16 MHz
- Flash Memory: 32 KB (of which 2 KB is used by the bootloader)
- SRAM: 2 KB
- EEPROM: 1 KB
- Digital I/O Pins: 14 (6 of which provide PWM output)
- Analog Input Pins: 6
- PWM Resolution: 8-bit
- ADC Resolution: 10-bit
- Communication Interfaces: UART, SPI, and I2C
- USB Interface: Mini USB Type-B
- Operating Current: 40 mA (per I/O pin maximum)
- DC Current for 3.3V Pin: 50 mA
- Power Consumption: Low (ideal for battery-powered applications)
- Dimensions: 45 mm x 18 mm
- Weight: Approximately 7 grams
- Supports ISP Programming: Yes (In-System Programming)
- Breadboard Friendly: Designed to fit directly on breadboards
- Open Source Hardware: Circuit design and software are freely available

ULTRASONIC SENSOR



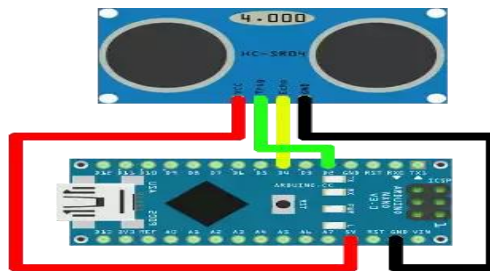
Fig 3.3 Ultrasonic Sensor

An ultrasonic sensor is a device used to measure the distance between the sensor and an object by using high-frequency sound waves. It operates on the principle of echolocation, similar to how bats detect obstacles. The sensor emits ultrasonic waves, typically at 40 kHz, from its transmitter. When these waves hit an object, they reflect back to the receiver. The time taken for the waves to return is measured, and the distance is calculated using the formula:

$$\text{Distance} = \frac{\text{Speed of Sound} \times \text{Time}}{2}$$

The most common type, the HC-SR04 ultrasonic sensor, consists of four pins — VCC, GND, TRIG, and ECHO. It can measure distances ranging from 2 cm to 400 cm with high accuracy. Ultrasonic sensors are widely used in robots, automated systems, vehicle parking sensors, and industrial applications for distance measurement, object detection, and collision avoidance.

Circuit Diagram of Ultrasonic Sensor With Arduino



Features

- Works on sound waves (around 40 kHz).
- Measures distance without touching the object.
- Range: 2 cm to 400 cm.
- Has four pins – VCC, Trig, Echo, and GND.
- Works with 5V power supply.
- Used for obstacle detection and distance measurement in robots and vehicles.

L298N Motor Driver

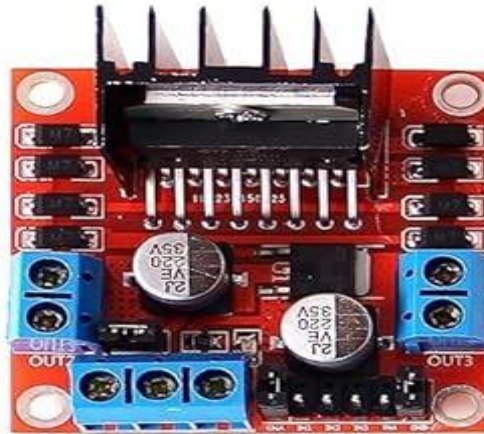


Fig 3.4 L298N Motor Driver

The L298N Motor Driver is a dual H-bridge motor driver module that allows bidirectional control of two DC motors or one stepper motor. It is widely used in robotics and embedded systems for controlling motor direction and speed using logic-level signals from microcontrollers like the Arduino Uno.

In the maze-solving robot project, the L298N serves as the interface between the Arduino and the DC motors. Since the Arduino cannot supply sufficient current to drive motors directly, the L298N amplifies the control signals received from the Arduino and provides the necessary voltage and current to the motors. It also supports Pulse Width Modulation (PWM), enabling speed control in addition to directional control.

The module typically includes:

- Two H-bridge circuits (for controlling two motors independently)
- Input pins for control signals (IN1, IN2, IN3, IN4)
- Enable pins for speed control (ENA, ENB)
- Output pins to motors (OUT1–OUT4)
- Power supply input (usually supports 5V–12V motor power)

The L298N is robust, easy to integrate, and capable of handling motor currents up to 2A per channel, making it ideal for small to medium-sized mobile robots. Its ability to control both motors independently is essential for maneuvering the robot accurately through the maze.

DC BO MOTOR

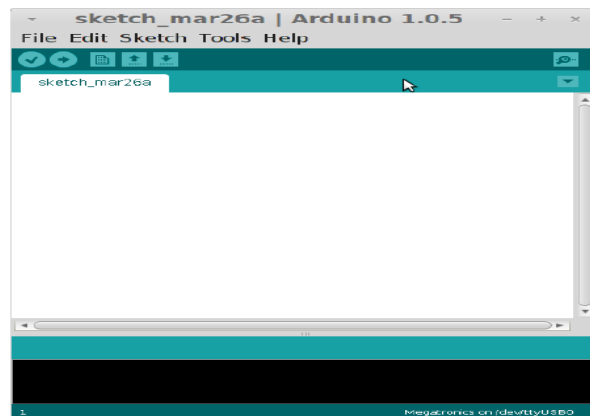


Fig 3.5 DC BO Motor

A DC BO motor (Battery Operated motor) is a compact and efficient geared motor designed to operate on low DC voltage, typically between 3V and 12V. It is widely used in robotics and embedded systems because it provides high torque at relatively low speed, making it ideal for driving wheels and mechanical components in small robots. The motor functions based on electromagnetic principles: when electrical current flows through the coil inside the motor, it generates a magnetic field that interacts with the permanent magnets, causing the shaft to rotate. What makes a BO motor especially useful is its built-in gearbox, which reduces rotational speed and increases torque output, allowing controlled and powerful motion suitable for applications like robotic vehicles.

The DC BO motor is lightweight, cost-effective and easy to mount on robotic chassis. However, since it draws more current than a microcontroller pin can supply, it cannot be connected directly to an Arduino or similar controller. Instead, a motor driver such as the L298N, L293D or TB6612FNG is used to control direction and speed, often through PWM signals. BO motors are available in different types, including straight-shaft, dual-shaft (for encoders) and offset-shaft versions to fit various mechanical designs. They are commonly used in line-following robots, maze-solving robots, smart car projects, mini conveyors and automatic systems. Their simplicity, reliability, high torque output and easy interfacing make them an excellent choice for beginners and advanced robotic applications.

ARDUINO IDE



Arduino Fig 3.6 IDE

Arduino is an open-source project, enabling hobbyists to easily take advantage of the powerful Atmega chips. The Arduino IDE is the software where you can write code and upload it to the Atmega chip. The code is then executed on the chip.

Most 3D-printer electronics are Arduino-compatible, they use the Atmega chip and enable the user to upload their code using Arduino. This includes Megatronics, Minitronics and RAMPS.

Before you can start using the electronics you need software 'firmware', that translates machine instructions (gcode) into actual movements. There are a few options here, including Marlin and Sprinter and Repetier. The actual firmware is not discussed in this document. You can use Arduino to upload this firmware onto your electronics. This document will guide you in the steps you need to take.

To upload a firmware, you must first open the files using File → Open. Select the .ino file from the directory containing the firmware. Arduino will open several tabs with files.

Next step is to select the correct electronics board. From the Tools menu, locate the Board item. This item should include a few sub items, including Megatronics, Minitronics, Arduino mega 2560 (RAMPS with mega 2560) and Arduino Mega 1280 (RAMPS with mega1280). Select the board that fits your electronics.

Also we need to select the serial port the electronics is connected to. In the Tools menu, locate the Serial port item. This should include at least one item if the board is connected and the drivers are installed properly. If there are multiple items here, you need to find out which is the correct one by unplugging the board and checking which port was removed.

Once you have set the board and serial port, you can upload the firmware by pressing File → Upload. Arduino will try to compile the firmware, if any errors occur the process will stop and you will need to fix the errors before trying again.

Once compilation is complete, the actual upload will start. This may take a minute for a large sketch. You can actually 'talk' to the firmware using the Serial monitor. Make sure the correct serial port is selected and locate the Serial monitor button.

HARDWARE IMPLEMENTATION



Fig 3.7 Hardware Implementation

Initialization and Setup

The initialization of a maze-solving robot using Arduino Nano includes connecting sensors to detect walls, motors via a driver and defining their pins in the program. In this sensors are set as inputs, motors as outputs and serial communication begins, preparing the robot for maze-solving execution.

Arduino Nano – Main controller for processing sensor input and controlling motors.

Ultrasonic Sensors – For wall/obstacle detection and maze path tracking.

Motor Driver (L298N) – Interface between Arduino and DC motors.

DC BO Motors with Wheels – For movement and navigation inside the maze.

Chassis – Base structure to mount all components.

Power Supply – To power Arduino and motors.

Chapter 4

RESULTS AND DISCUSSIONS

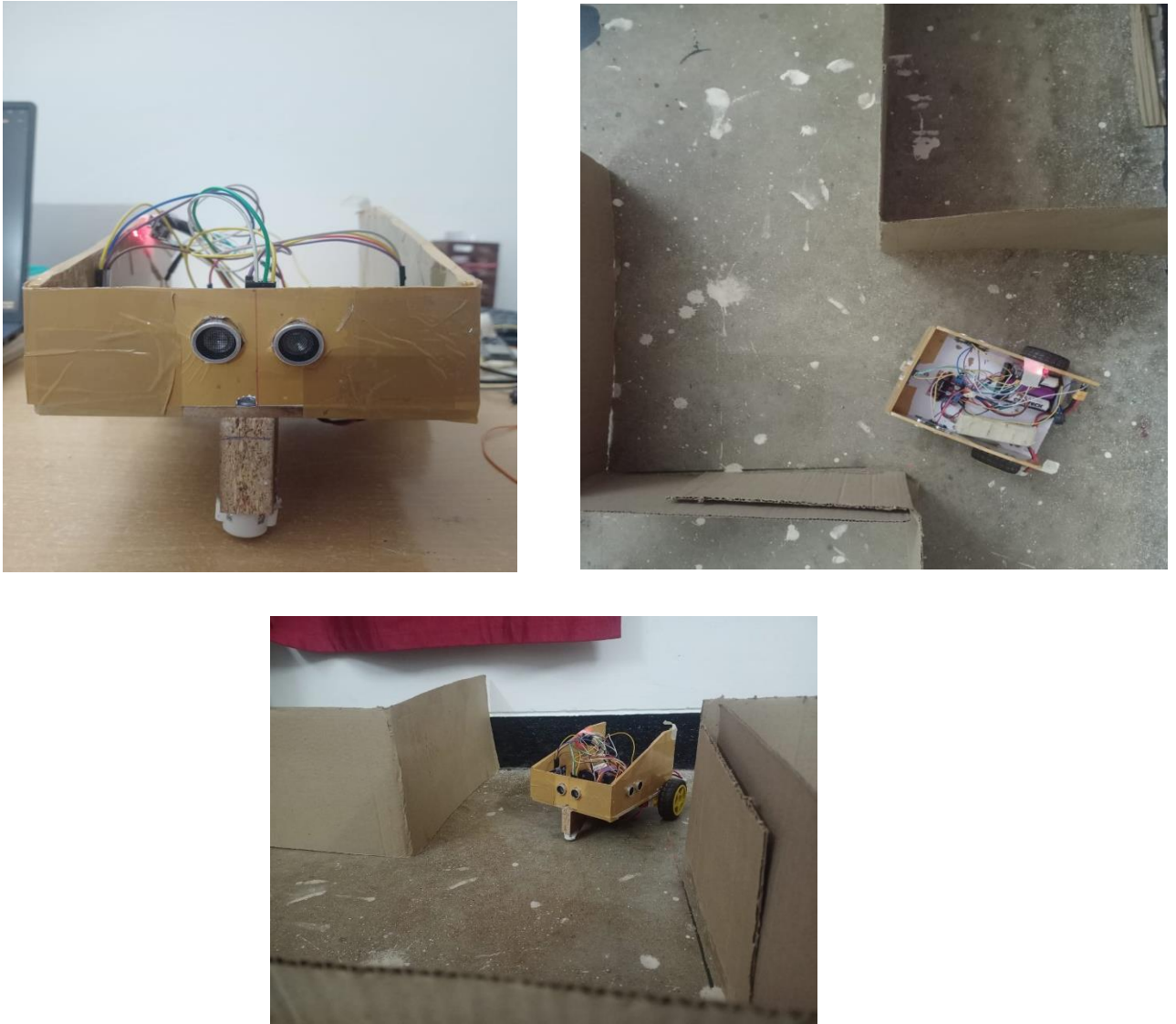


Fig 4.1 Result

The maze-solving robot was successfully designed and implemented to navigate through a given maze using sensor feedback and pathfinding algorithms. The robot was able to detect paths, avoid dead ends, and reach the destination autonomously. It demonstrated accurate decision-making in selecting routes and achieved reliable performance in solving different maze patterns. The project validates the effectiveness of using microcontrollers, sensors, and algorithms in developing autonomous navigation systems.

4.2 ADVANTAGES

- Autonomous Navigation
- Problem-Solving Ability
- Efficient Pathfinding
- Human Risk Reduction
- Foundation for AI & Robotics Research

4.3 DISADVANTAGES

- High Cost of Development
- Complex Programming
- Limited Battery Life
- Slow Processing in Complex Mazes

4.4 APPLICATIONS

- Disaster Rescue Operations.
- Military and Defense.
- Space Exploration.
- Search and Exploration.
- Warehouse & Industrial Automation.
- Medical Applications.
- Education & Research.
- Service Robots

CONCLUSION

The maze-solving robot demonstrates the integration of robotics, sensors, and intelligent algorithms to achieve autonomous navigation. By solving mazes and finding optimized paths, the robot proves its potential in real-world applications such as disaster rescue, defense, industrial automation, and education. Despite limitations like sensor accuracy and power constraints, the project provides a strong foundation for understanding autonomous systems, pathfinding techniques, and robotics research. Overall, the maze-solving robot is an excellent prototype that highlights the possibilities of artificial intelligence and robotics in solving complex navigation problems.

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